

Touchless Web Portal System Overview

The PecoDrop Touchless Web Portal provides a complete 100% touchless mobile interface for corporate employees and citizens operating Reverse Vending Machines (RVM). Users scan a dynamic QR code displayed on the kiosk, authenticate via Google SSO or Mobile Number, trigger physical door unlocks, monitor live container telemetry in real-time via the newly updated Live Progress card, and settle earned rewards directly into their Eco Wallet.

■ Updated Touchless 3-Screen Operational Workflow

SCREEN 1: Start Recycling Mode

- QR Code Scan Landing Page
- Google SSO & Corporate Auto-Detect
- Mobile Phone Number Input
- Kiosk Remote Start Trigger Button

SCREEN 2: Updated Kiosk Started

- Kiosk Aperture Unlocked Banner
- **NEW Live Kiosk Progress Card**
- Live Split Containers & Points
- Integrated Finish & Claim Button

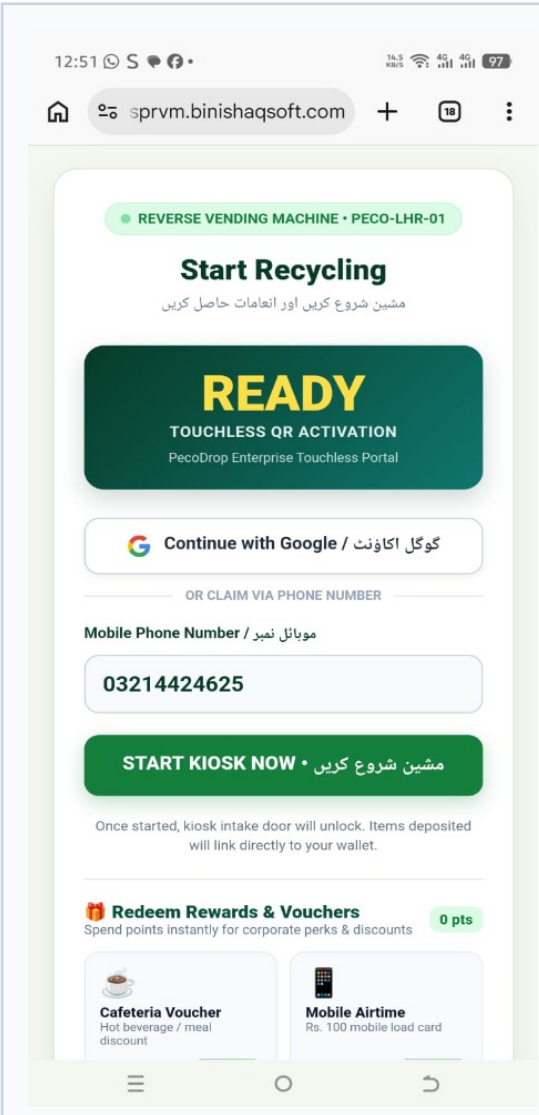
SCREEN 3: Points Claimed Mode

- Session Completion Celebration
- Eco Wallet Points Credit (+35 PTS)
- Available Points Balance Sync
- Touchless Voucher & Perk Store

■■ Key Architectural Highlights

- **Zero Physical Contact:** Scanning dynamic QR code opens smartphone portal, bypassing kiosk touchscreen and physical keypad.
- **Updated Live Telemetry Card:** Displays split real-time CONTAINERS and EARNED +PTS with embedded FINISH & CLAIM button.
- **Single Source of Truth:** Real-time telemetry, session state, and wallet balances write exclusively to **PostgreSQL (rvmpg)**.
- **Browser Hygiene & Safety:** window.history.replaceState purges startToken query parameters upon session finish, preventing stale tab re-locks.
- **Dual Desktop Compatibility:** Native hardware integration with both **PecoDropDesktopApp** (Enterprise Corporate) and **RVMDesktopApp** (Public).

■ SCREEN 1: Start Recycling & Touchless QR Authentication Mode



Screen 1 Component & Authentication Breakdown:

1. Kiosk Identification Header:

Displays machine badge REVERSE VENDING MACHINE • PECO-LHR-01
confirming active QR handshake target.

2. Action Title & Urdu Subtitle:

Displays **Start Recycling** (■■■■ ■■■■ ■■■■ ■■■ ■■■■■■■■ ■■■■ ■■■■) inviting immediate touchless activation.

3. System Status Badge:

Prominently displays **READY - TOUCHLESS QR ACTIVATION** indicating kiosk is idle and ready to initiate session.

4. Corporate Google SSO Authentication:

1-Tap **Continue with Google** /  button. Corporate email domains (@engro.com, @bankalfalah.com) automatically map user to enterprise perks and departments.

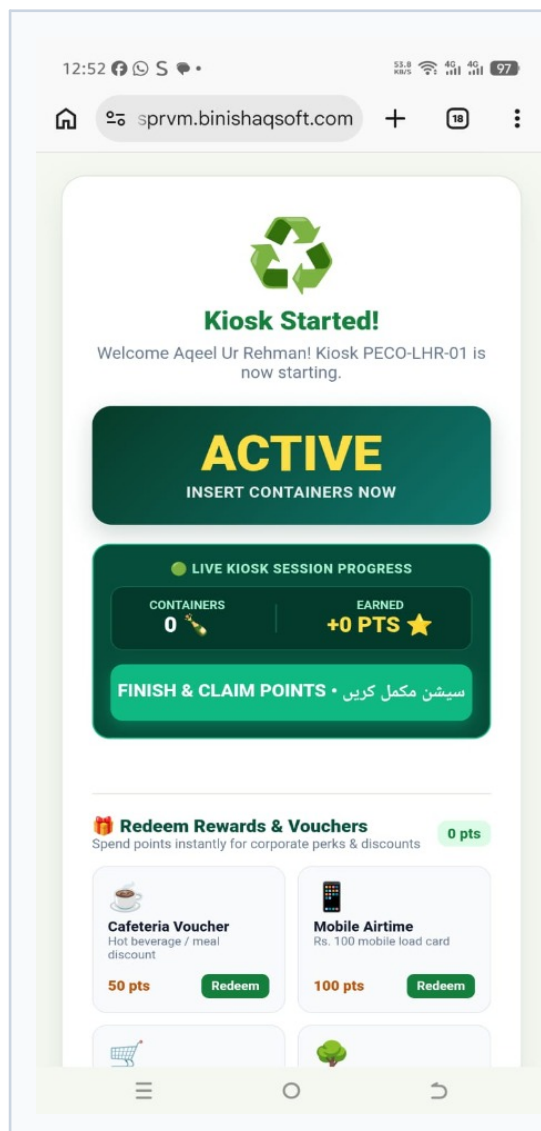
5. Mobile Phone Number Option:

Alternative direct phone entry field (e.g. 03214424625) for quick guest or non-SSO authentication.

6. Hardware Start Trigger Button:

Prominent **START KIOSK NOW** • ■■■■ ■■■■ ■■■■ green button sends immediate handshake command to unlock hardware apertures.

■ SCREEN 2: Updated Kiosk Started & Live Session Progress Mode



Screen 2 Component & Live Telemetry Breakdown (Updated UI):

1. Authenticated Greeting & Machine Banner:

Displays **Kiosk Started!** with green recycling logo and personalized greeting: *Welcome Aqeel Ur Rehman! Kiosk PECO-LHR-01 is now starting.*

2. Unlocked Hardware Status Banner:

Prominently displays **ACTIVE - INSERT CONTAINERS NOW** green card confirming physical aperture doors (■ Plastic, ■ Can, ■ Paper) are open.

3. NEW Live Kiosk Session Progress Widget:

Features dark emerald card with ● LIVE KIOSK SESSION PROGRESS header, streaming live container intake metrics updated every 1.5 seconds:

- **CONTAINERS:** Live count of deposited items (0 ■)
- **EARNED:** Real-time points calculation (+0 PTS ■)

4. Integrated Remote Session Finish Button:

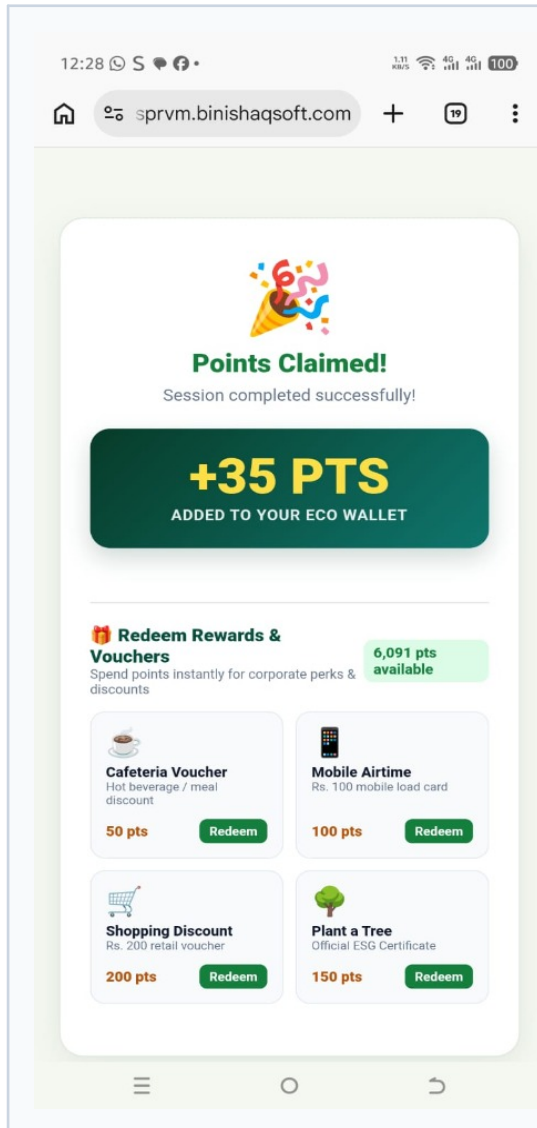
Embedded directly inside the Live Progress Card: **FINISH & CLAIM POINTS**

- ■■■■ ■■■■ ■■■■ button for 1-tap remote session settlement.

5. Touchless Rewards Catalog Teaser:

Displays current user reward balance (0 pts) and quick access to corporate perks (Cafeteria Vouchers, Mobile Airtime, Retail Vouchers, ESG Certificates).

■ SCREEN 3: Points Claimed & Eco Wallet Settlement Mode



Screen 3 Component & Settlement Breakdown:

1. Session Celebration Header:

Displays celebration icon and confirmation message: *Points Claimed! Session completed successfully!*

2. Earned Points Credit Banner:

Prominently displays earned session reward: **+35 PTS - ADDED TO YOUR ECO WALLET** credited atomically in PostgreSQL database.

3. Total Eco Wallet Balance Badge:

Updates and displays user total accumulated points balance in real-time (**6,091 pts available**).

4. Browser History Cleaning (replaceState):

Executes `window.history.replaceState` to automatically strip `startToken` query parameter from browser address bar, preventing stale tab re-locks.

5. Touchless Reward Redemption Store:

Allows instant point redemptions for **Cafeteria Voucher** (50 pts), **Mobile Airtime** (100 pts), **Shopping Discount** (200 pts), and **Plant a Tree** ESG certificate (150 pts).

■ System Capabilities & Technical Architecture

1. Dual Desktop Engine Compatibility:

The Touchless Web Portal communicates seamlessly with both **PecoDropDesktopApp** (Enterprise Campus Kiosks with dual landscape displays & paper weight load cell) and **RVMDesktopApp** (General Public Machines). Both desktop apps poll the central handshake endpoint and execute atomic wallet settlements.

2. Single Source of Truth — PostgreSQL (rvmpg):

All session transactions, user point balances, voucher redemptions, machine statistics, and organizational departments write exclusively to PostgreSQL. Legacy MongoDB (`rvmapg`) is strictly read-only for historical sync.

3. Automatic Corporate SSO Domain Verification:

Email domains matching registered corporate organizations (e.g. `@engro.com`, `@bankalfalah.com`, `@ucp.edu.pk`) are automatically provisioned as **ENTERPRISE** users, linking department metrics and cafeteria perk discounts.

4. Browser History Cleaning (`replaceState`):

Upon session completion, the web portal automatically executes `window.history.replaceState` to clean query string tokens. Closing, refreshing, or re-opening the browser tab lands cleanly on the **Start Recycling** screen without locking into stale completed states.

5. Touchless Reward Redemption Store:

Users can spend earned points instantly for Cafeteria Meal Vouchers, Mobile Airtime Cards (EasyPaisa / JazzCash), Retail Discounts, or ESG Tree Certificates directly from the smartphone portal.